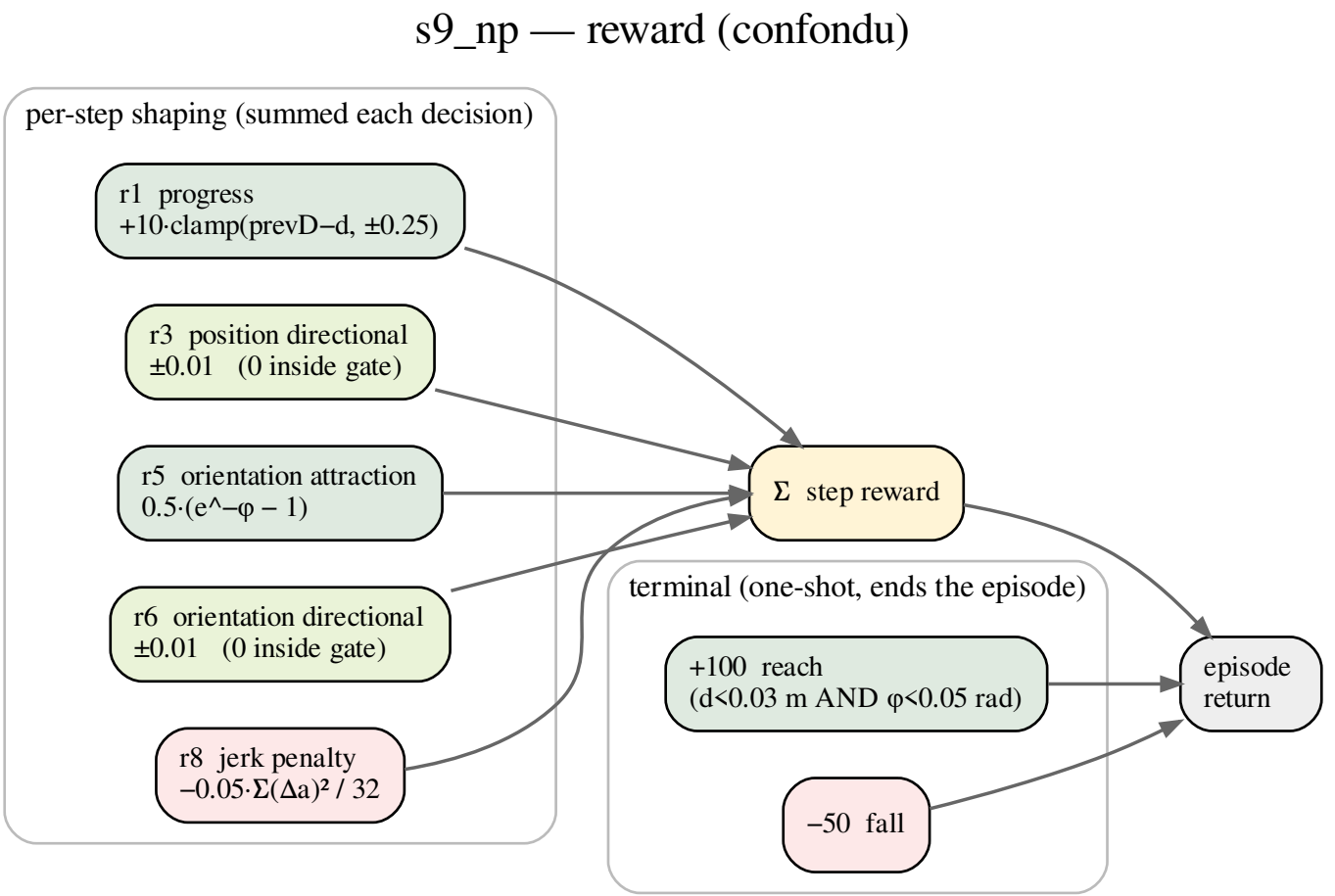
s9_np — reward (confondu)
per-step shaping (summed each decision)
r1 progress
+10·clamp(prevD−d, ±0.25)
r3 position directional
±0.01 (0 inside gate)
r5 orientation attraction
0.5·(e^−φ − 1)
r6 orientation directional
±0.01 (0 inside gate)
r8 jerk penalty
−0.05·Σ(Δa)² / 32
Σ step reward
terminal (one-shot, ends the episode)
+100 reach
(d<0.03 m AND φ<0.05 rad)
−50 fall
episode
return